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Epidemiological status of kissing-bugs in South East Asia: A preliminary assessment

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ABSTRACT

Kissing-bugs (Triatominae) are being increasingly reported as a biting nuisance in SE Asia, with severe bite reactions sometimes leading to anaphylactic shock. In addition, they pose a risk for vector-borne transmission of trypanosomiasis, with potential diagnostic difficulties due to the range of trypanosome species in the region. Here, we review available information about Triatominae in Asia, and present additional comparisons using morphometry, cytogenetics, and new DNA sequence data, to clarify their relationship with each other and with the better known American species. We deduce that all Asian Triatominae have probably derived from forms originally spread during the 15–18th centuries on sailing ships, from the area that now forms the southern USA.

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1. Introduction

Most species of kissing-bug (Hemiptera, Reduviidae, Triatominae) occur in America, where, they are well-known as vectors of *Trypanosoma cruzi* – causative agent of Chagas disease (American trypanosomiasis). This infection is not yet recorded from SE Asia – except in previously-infected immigrants from America – but the kissing-bugs themselves are of increasing nuisance in many SE Asian cities. At present, the main nuisance appears to be due to bite reactions – which are sometimes serious enough to give fever, and in at least one case resulted in death attributed to anaphylactic shock (see Dujardin et al., 2015). The most widespread of these kissing-bugs, *Triatoma rubrofasciata*, is a competent vector of *T. cruzi* in Latin America, and is also frequently infected with a rat trypanosome – *Trypanosoma conorhini* – which is not yet known to infect other domestic animals or humans [but see Deane et al., 1986]. Kissing-bugs may also be capable of transmitting other trypanosomes, such as *Trypanosoma lewisi* which is able to infect humans – with severe consequences.

In epidemiological terms therefore, kissing-bugs in Asia seem to pose an actual problem due to bite reactions that can be serious enough to require treatment, and a potential problem as vectors of trypanosome parasites to humans and/or domestic animals. In addition, since little is known about the range of trypanosomes that may be able to infect Asian Triatominae, it is possible that some of these may emerge as human pathogens, and several may provide a source of confusion in parasite diagnosis. Here, we review what is currently known and surmised about kissing-bugs and their parasites in SE Asia, and reassess the questions surrounding their origins.

2. The insects

The Triatominae are represented in Asia by two genera: *Linshcosteus* with six species described from India, and *Triatoma* with eight species described from various points within the area from India to southern China and northern Australia. The most widespread of these, *T. rubrofasciata*, is also recorded from port areas in Africa and the Americas, and from various islands in the Caribbean, Pacific and Indian oceans; it was the first species of

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Table 1
Type depositories for Asiatic Triatominae.

AMNH	American Museum of Natural History, New York, USA
BMH	Bishop Museum, Honolulu, Hawaii
DEI	Deutsches Entomologisches Institut, Müncheberg, Germany
FIOCRUZ	Instituto Oswaldo Cruz, Rio de Janeiro, Brasil
IRSNB	Institut Royal des Sciences Naturelles de Belgique, Brussels, Belgium
MNHN	Muséum National d'Histoire Naturelle, Paris, France
NHM	Natural History Museum, London, UK
NRS	Naturhistoriska Riksmuseet, Stockholm, Sweden
NUC	Nankai University Collection, Tientsin, China
RNH	Rijksmuseum van Natuurlijke Historie, Leiden, The Netherlands

kissing-bug to be described, from specimens encountered "aux Indes" (De Geer, 1773), and remains the type species of the genus and subfamily. Type depositories are indicated in Table 1.

2.1. Linshcosteus Distant

L. carnifex Distant, 1904. (holotype: NHM) This species is known only from the female holotype; it bears a label reading "E. India" but Distant (1904) indicates the specimen to be from north India.

L. chota Lent and Wygodzinsky, 1979. (holotype: NHM) This species is known only from the male holotype; the specimen is from south India and was previously labelled as the allotype of *L. carnifex*.

L. confumus Ghauri (1976). (holotype + paratypes: NHM) is said to be widespread in south India (Mysore) where it has been collected (as *L. carnifex*) from under rocks (Sankaran and Nagaraja, 1975).

L. costalis Ghauri (1976). (holotype + paratypes: NHM) seems to be allopatric with *L. confumus*, and to share the same microhabitat (Ghauri, 1976).

L. kali Lent and Wygodzinsky (1979). (holotype + paratypes, FIOCRUZ; others: NHM, AMNH) This species has been recorded from several localities of south India, including Madras state (Lent and Wygodzinsky, 1979).

L. karupus Galvão et al. (2002). (holotype + paratypes: FIOCRUZ; others NHM) This species was described from dark-coloured specimens collected amongst rockpiles near Kalakkadu, Tamil Nadu state, southern India (Galvão et al., 2002).

2.2. *Triatoma* Laporte

T. amicitiae Lent (1951). (holotype: NHM) This species is known only from the female holotype, believed to have been collected in southern Sri Lanka (Monteith, 1974). It is the smallest of the Old World *Triatoma* (Lent and Wygodzinsky, 1979)

T. bouvieri Larrousse (1924). (holotype: MNHN) This species was described from Nha-Trang in southern Vietnam (Larrousse 1924) and subsequently said also to occur in the Nicobar islands (Lent, 1951) and Philippines (Lent and Wygodzinsky, 1979) [the specimen in the BMH collection is labelled as collected at the mouth of the Malabanga River, Palawan, Philippines]. It is very similar to *T. migrans* by morphology (Lent and Wygodzinsky, 1979) and to migrans and rubrofasciata by morphometry (Gorla et al., 1997).

T. cavernicola Else et al. (1977). (paratypes: NHM) This species is described only from limestone caves near Perlis in northern Malaysia, where it is believed to feed from bats and occasionally to attack humans (Else et al., 1977).

T. leopoldi (Schoudeten, 1933) (=*novaeguineae* Miller (1958)) (holotypes: *leopoldi*: IRSNB; *novaeguineae*: RNH) *T. leopoldi* was described from two specimens from West Irian, Indonesia (from Sakoemi and Wemdessi) (Schoudeten, 1933), and said also to have been found in Sulawesi (Lent, 1951). *T. novaeguineae* was described from two specimens from Biak Island off the coast of West Irian (Miller, 1958); another specimen was collected in Papua New Guinea, and three additional male specimens taken by light trap at Iron Range, on the Cape York peninsula of northern Queensland, Australia – close to some abandoned mine shafts (Monteith, 1974). The two were synonymised by Lent and Wygodzinsky (1979).

T. migrans Breddin 1903 (=*pallidula* Miller 1941) (holotypes: *migrans*: DEI; *pallidula*: NHM; others: BMH). This species has been recorded from localities in Indonesia (Borneo, Java, Sumatra), Malaysia (Sarawak), Philippines (Mindanao), and Thailand. A specimen from Sarawak was collected in "rotten heartwood of a felled tree" (Lent and Wygodzinsky, 1979).

T. pugasi Lent 1953. (holotype: NHM). This species was described from a single male from Indonesia (Java) in the NHM collection. A female is described and photographed in Lent and Wygodzinsky (1979) but without any further details of where the specimen was collected, nor where it is deposited.

T. rubrofasciata (De Geer, 1773) (=*Conorhinus stalii* Signoret, 1860) (holotypes: *rubro-fasciatus* and *stalii*: NRS ; others : AMNH, FIOCRUZ, NHM, BMH). This species is very widely recorded from port areas throughout the Indo-Pacific region, including southern China and Taiwan; it is also recorded from a few port areas of Africa and the Caribbean, and from the Atlantic port areas of Florida and Latin America (see: Ryckman and Archbold, 1981; Schofield et al., 2009). It was the first of the Triatominae to be described, as "*Cimex rubro-fasciatus*" (De Geer, 1773), from one (or more) specimens said to be found "aux Indes" (assumed to refer to the then Dutch East Indies – i.e., Indonesia, or possibly Indochina). Being widespread and morphologically very variable, it has acquired at least 11 synonyms (see Lent and Wygodzinsky, 1979) including *variegatus* Stal (1872); which is also a synonym for *T. sanguisuga* (Leconte 1855) (part) and *T. lecticularia* Stal 1859 (part) (Stal, 1872; Lent and Wygodzinsky, 1979). *T. rubrofasciata* seems strongly associated with domestic and peridomestic habitats – especially with rats (especially *Rattus rattus*) – and is frequently infected with the rat trypanosome *T. conorhini*. It is also competent as a vector of *T. cruzi*, and considered to be the most significant vector of Chagas disease on the Island of São Luis de Maranhão, Brazil (Rebêlo et al., 1998).

T. sinica Hsaio 1965 (holotype: NUC; female paratype: AMNH). This little-known species was described from specimens collected in Nanking, China.

3. The parasites

The main parasites transmitted by kissing-bugs are trypanosomes (Protozoa, Kinetoplastida, Trypanosomatidae). The Trypanosomatidae appear to be a monophyletic group forming three main clades (Stevens et al., 2001). The "aquatic clade" includes species that parasitise fish and amphibians, probably mainly transmitted by leeches. The "African clade" mainly includes species that are cyclically transmitted by tsetse-flies (Glossinidae) or mechanically transmitted by other blood-sucking flies such as horseflies (Tabanidae) or stableflies (Stomoxinae); the latter group includes *Trypanosoma evansi*, well-known as causative agent of "surra" – an often fatal disease of horses and camels that is now widespread in parts of SE Asia, and can cause serious infections in people (e.g., Shegokar et al., 2006; Vanhollebeke et al., 2006). Also in this clade, *Trypanosoma vivax* is a widespread parasite of cattle that has spread into SE Asia, transmitted by biting flies.

The third of these main clades, sometimes known as the "*cruzi* clade", includes trypanosomes related to *T. cruzi* – causative agent of American trypanosomiasis (Chagas disease). Amongst these are the so-called "megatrypanum" species such as *Trypanosoma conorhini* transmitted by Triatominae, bird-infective species such as *Trypanosoma avium* transmitted by mosquitoes or simuliid

blackflies (e.g., Votýpka et al., 2002), bat-infective species such as *Trypanosoma dionisii* and *T. vespertilionis* transmitted by cimicid bedbugs (see: Gardner and Molyneux, 1988; Barnabé et al., 2003), and the widespread *T. lewisi* normally transmitted between rodents by fleas, but which can also infect primates (Maia da Silva et al., 2010) and humans – sometimes with fatal outcome (Kaur et al., 2007; Banerjee et al., 2008; Verma et al., 2011).

4. Epidemiological aspects

Trypanosomiasis is well-known in the Americas as Chagas disease due to infection with *T. cruzi*. Similarly in Africa, several trypanosomes are of medical and veterinary significance, including forms of *Trypanosoma brucei* causing "sleeping sickness" in humans, and other species of importance as livestock pathogens such as *Trypanosoma brucei*, *T. congolense*, *T. vivax*, and *T. simiae*. In Asia however, trypanosomes have been of minor importance as human and livestock pathogens, although there is increasing evidence of serious infections with a surprising range of trypanosome parasites in the region. As mentioned above, both *Trypanosoma vivax* and *T. evansi* seem to be spreading as parasites of cattle and other domestic animals, and fatal human infections with *Trypanosoma evansi* and *T. lewisi* have recently been reported from the Asian region – mainly from the Indian subcontinent. In parallel, there is increasing evidence for a spread of *T. cruzi* to the region, mainly carried by asymptomatic immigrants from the Americas (e.g., Coura and Viñas, 2010). To us, these features indicate two main epidemiological challenges. First is the evident risk of vector-borne transmission of *T. cruzi* commencing in Asia. But second, are the diagnostic difficulties that may be posed by parasitological confusion between similar trypanosomatid forms seen in the bloodstream of infected humans or domestic animals, or in the intestinal contents of infected vectors, and the serological cross-reactions that may (or may not) occur between *T. cruzi* and widespread Asian trypanosomes of the same clade such as *Trypanosoma conorhini* or *T. lewisi*. As a further complication, we can see no obvious impediment to the future spread of New World vectors to the Old World (see Schofield et al., 2009), just as it appears to have occurred in the relatively recent past (see below).

There is an additional puzzle relating to *Trypanosoma conorhini*, which is a highly prevalent infection of *T. rubrofasciata* in both Old and New World populations – often reaching well over 50% infection rates amongst the older bug stages. A similar parasite has also been found in *Linshcosteus* collected in southern India (Miles, 2013). *Trypanosoma conorhini* is known to infect rats, but generally seems to develop only transient infections (Hoare, 1972). By contrast, the trypanosomes reported from naturally infected macaques in Taiwan (Cross et al., 1983) and Indonesia (Weinman and Wiratmadja, 1969) are believed to be *Trypanosoma conorhini* (Hoare, 1972; Deane et al., 1986) suggesting that primates may be an important reservoir for this parasite. So if *Trypanosoma conorhini* is – or becomes – capable of infecting humans, this would further complicate the epidemiological pattern of trypanosomiasis in Asia.

At present however, the main epidemiological problem posed by Triatominae in Asia is the severe reaction that may develop from the bite. Even earliest reports describe the bite of *T. rubrofasciata* as "more dangerous than the scorpion" and followed by "a tumor the size of a pigeon's egg" (for references, see Dujardin et al., 2015). Today, bite reactions to *T. rubrofasciata* are mainly reported from parts of southern China, Philippines, and Vietnam. People bitten present swelling and itching at the site of the bite, sometimes with consecutive local skin infection. In Hanoi, 24% of cases also involved severe fever lasting 1–2 days (Trung Xuan and Dujardin, 2013) although for the whole of Vietnam the average fever rate following a bite was around 5% (Van Chau et al., 2013). Severe immunological reactions have also been reported, including cases of fatal anaphylaxis in Singapore and China (Teo and Cheah, 1973; Wang and Peng, 2006).

5. Evolutionary aspects

The presence of kissing-bugs in Asia has long attracted speculation about their origin, with several authors suggesting that the Old World species represent the original forms. It had been suggested that the subfamily as a whole was of Old World origin, that had either migrated to the Americas or represented an ancient worldwide group that had since died out in many areas. (Ryckman and Archbold, 1981). Neither of these ideas is easy to sustain however, because the New World is clearly the site of greatest diversity of the Triatominae. And although silvatic populations of Triatominae are widespread in the Americas, from about 42°N to 46°S, there are no silvatic populations of Triatominae reported from Africa, central Asia, or the Middle East.

It had also been suggested that the two genera represented in the Old World – *Triatoma* and *Linshcosteus* may have had different origins (e.g., Schaefer and Coscarón, 2001). Species of *Linshcosteus* differ notably from Old World *Triatoma* by their broad abdomen and short wings (similar to Mexican species of the *Triatoma phyllosoma* complex) and also by their truncated rostrum that does not reach the prosternal sulcus (a feature shared only with the Cavernicolini). The idea that *Linshcosteus* might have arisen independently was formalised by creation of the tribe "Linshcosteusini" (sic) (Carcavallo et al., 2000), but is not supported by genetic comparisons (e.g., Hypsa et al., 2002; Patterson et al., 2001; Schofield, 2004) which usually cluster *Linshcosteus* with *T. rubrofasciata*. Nor is it supported by morphometric analyses based on the wing venation patterns (see below). In spite of their other morphological differences, the wing venation of *Linshcosteus* presents a geometry that is very close to that of *T. rubrofasciata*, close also to that of North American species of *Triatoma*, but quite distinct from that of South American *Triatoma* (see below).

Current consensus is that the Triatominae are of relatively recent origins in the Americas (see: Schofield and Galvão, 2009; Hwang and Weirauch, 2012) and that the Old World species represent derivatives from an American form. Noting its close association with rats, especially ship-rats (*Rattus rattus*), and its occurrence in port areas throughout the tropics and subtropics, the idea is that *T. rubrofasciata* was accidentally carried on ships from the Americas to elsewhere. Under that hypothesis, *T. rubrofasciata* should show evidence of a close relationship with at least some of the New World species, and all the Asian Triatominae – including the morphologically distinct *Linshcosteus* – would all appear to be derivatives of *T. rubrofasciata*.

6. Recent entomological data

To explore these hypotheses about the Asiatic Triatominae, we have made use of various techniques, including geometric morphometrics, cytogenetics and DNA sequencing of 28S, COI and Cyt-b gene fragments.

6.1. Geometric morphometrics

The insects used for morphometric analyses were dry, pinned, museum specimens, obtained from the Natural History Museum (NHM) (London, UK) [formerly British Museum of Natural History] and from the Royal Belgian Institute of Natural Sciences (IRSNB) (Brussels, Belgium). For this analysis, South America Triatomini were represented by *T. sordida* ($n = 11$), *T. eratyrusiformis* ($n = 2$), *T. maculata* ($n = 8$) and *T. pseudomaculata* ($n = 3$); Rhodniini were rep-

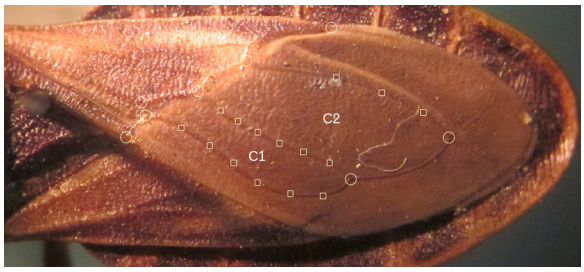

Fig. 1. Landmarks (circles) and semilandmarks (squares) on the wing of *T. rubrofasciata*. The two large internal cells of the wing are called C1 and C2: their complete contour has also been digitized (notshown).

resented by *Rhodnius sp.* ($n = 6$) and *Psammolestes sp.* ($n = 4$); North American Triatomini were represented by *T. sanguisuga* ($n = 7$), *T. lecticularia* ($n = 3$), *T. barberi* ($n = 3$) and *T. rubida* ($n = 2$). The Asiatic species of *Triatoma* included *T. bouvieri* ($n = 11$), *T. cavernicola* (n = 3), *T. leopoldi* ($n = 3$) and *T. migrans* ($n = 11$). *Linshcosteus* was represented by *L. karupus* ($n = 2$), *L. confumus* ($n = 4$), and *L. costalis* ($n = 2$). *Triatoma rubrofasciata* was represented by 45 specimens originating from all continents. Finally, four genera of predators belonging to other reduviid subfamilies were included: *Ectrychotes sp.* ($n = 8$), *Zelurus femoralis* ($n = 5$), *Reduvius sp.* ($n = 13$) and *Opisthacidius sp.* (n = 5).

Seven landmarks were selected on the wings, and 15 semilandmarks were used to capture the curved lines of veins between landmarks; contours were also taken of the two large internal cells of the wing, and analyzed separately (Fig. 1). All landmarks and semilandmarks were submitted to partial Procrustes superimposition, and semilandmarks then subjected to a sliding procedure. The tangent space projections were used as input for a principal component analysis and an UPGMA tree was derived from the Euclidean distances between groups (Fig. 3). Contours were submitted to elliptic Fourier analysis (EFA). The average contour was used to illustrate morphological variation between North and South American *Triatoma*, *T. rubrofasciata* and *Linshcosteus* sp. (Fig. 2). Landmark or contour digitization and subsequent morphometric analyses were performed using the CLIC package (http://momeclic.com) and the R software (http://cran.r-project.org).

The contours of internal cells C1 and C2 (Fig. 1) of the wing were clearly different between the South American and North American *Triatoma*, and *T. rubrofasciata* and *Linshcosteus* sp. both presented internal cell contours similar to those of North American *Triatoma* (Fig. 2).

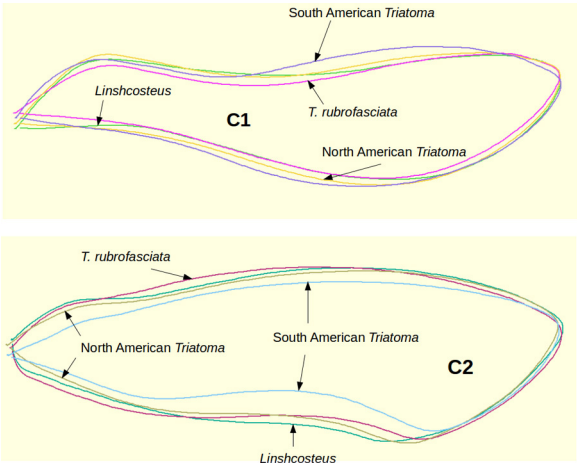

Fig. 2. Visualization of the C1 and C2 cells as superimposed after elliptic Fourier analysis. The South American *Triatoma* are showing a different contour.

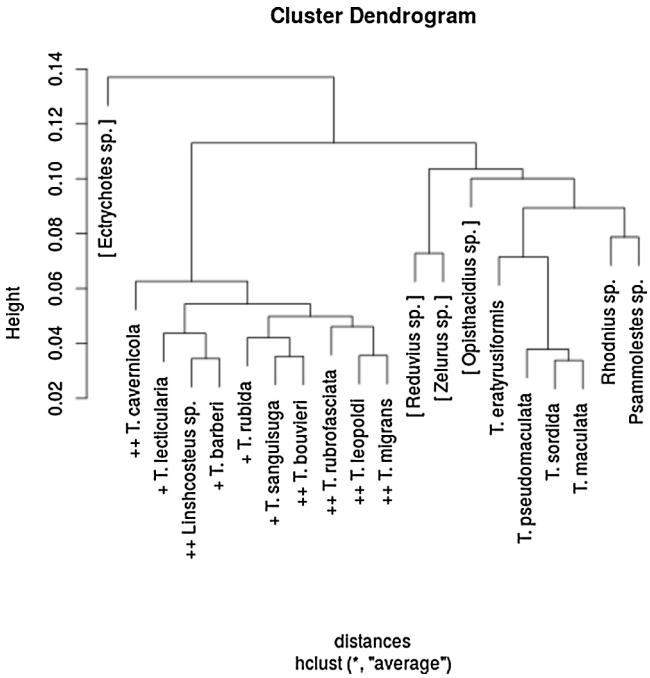

Fig. 3. UPGMA tree inferred from landmark-based shape variation, without any phylogenetic algorithm. "T.", *Triatoma*. "++", *Triatoma* from Asia. "+", *Triatoma* from North America. Names between square brackets are Reduviidae belonging to other subfamilies than Triatominae. Names ending by "sp." are genera.

The UPGMA tree derived from the principal components of wing shape (Fig. 3) reveals (i) complete separation between North and South American *Triatoma*, (ii) intimate mixing of the genus *Linshcosteus* with Asiatic and North American *Triatoma*, (iii) high similarity between the *Rhodnius* and *Psammolestes* genera, and (iv) the greater similarity of *Zelurus* clade members (as well as of the *Rhodnius–Psammolestes* pair) with South American Triatominae than with Northern and Asiatic *Triatoma*.

6.2. Cytogenetics and DNA sequencing

Cytogenetic observations were made on eight specimens of *T. rubrofasciata*: four specimens from Vietnam obtained from recent domestic captures in Hanoi, and four specimens from Brazil (2 from São Luis de Maranhão and 2 from Belem, Pará). The same specimens were also examined for nuclear and mitochondrial DNA variation.

For the cytogenetic studies, testes were removed from freshly killed adults, fixed in an ethanol–acetic acid mixture (3:1) and stored at −20 °C. Chromosomal techniques were applied using squashed gonad preparations made on slides in a drop of 50% acetic acid, freezing them in liquid nitrogen and removing the coverslip with a razor blade. The C-banding technique was performed as reported by Panzera et al. (1992) in order to determine the chromosome number and the distribution and behavior of C-heterochromatin during mitosis and meiosis.

Fluorescence in situ Hybridization (FISH) was carried out on 4 male *T. rubrofasciata* from Vietnam according to Panzera et al. (2012) in order to determine the chromosomal location of 45S ribosomal DNA clusters. Slides were analyzed using a Nikon Eclipse 80i epifluorescence microscope. Images were obtained with a Nikon DS-5Mc-U2 digital cooled camera using Nikon Nis Elements 3.1 Advanced Research software, and processed with Adobe Photoshop® software.

For DNA extraction, three legs fixed in 70% ethanol were used from each specimen. Total DNA was isolated by standard phenol–chloroform techniques, resuspended in 50 ml and stored

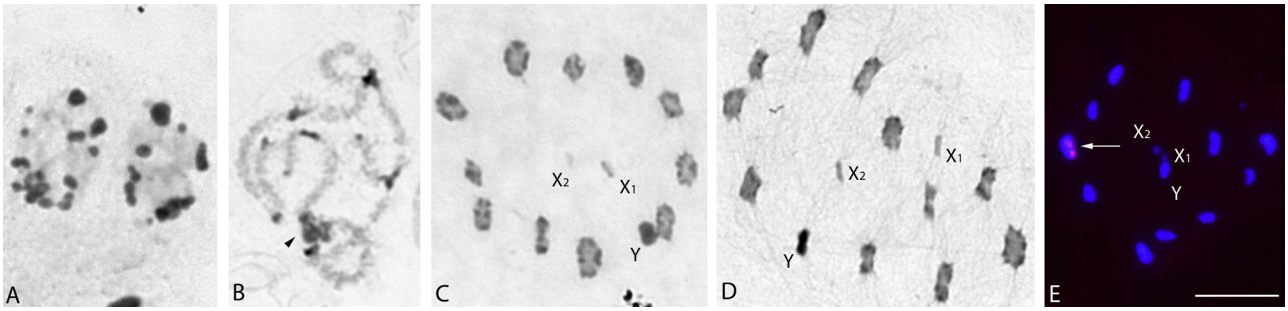

Fig. 4. Male meiosis of *Triatomarubrofasciata* from Brazil and Vietnam: 2n = 22 A + X1 × 2Y (males). (A–D): C-banding. (E): Fluorescent in situ hybridization with 45S ribosomal DNA probe. (A): Early meiotic prophases. Several C-heterochromatic chromocenters are observed scattered throughout the nuclei. (B): Pachytene. Autosomal terminal C-blocks are clearly observed in all autosomes. A largest chromocenter is constituted by the associated sex chromosomes (arrowhead). (C): First meiotic metaphase of an individual from Brazil. The eleven autosomal bivalents show heterochromatic C-blocks in both chromosomal ends. The Y chromosome is C-heterochromatic and the X2 chromosome is very smaller. (D): Metaphase I of an individual from Vietnam. Similar to (C) but the size of X2 chromosome is much larger. (E): Metaphase II of an individual from Vietnam. The 45S rDNA signals are located in one of the largest autosomal bivalent. Bar = 10μm.

at −20 °C until use. For each specimen, a cyt-b 667-bp fragment, COI 606-bp fragment, and a 28S rDNA 692-bp fragment were PCR amplified following Monteiro et al. (2000) for cytb and 28S rDNA and Calleros et al. (2010) for COI. PCR products were sent to Macrogen Inc., Korea, for sequencing. For all samples, sequencing was conducted both in forward and reverse directions.

For phylogenetic analyses, we used cyt-b and 28S rDNA sequences of other triatomine species deposited in GenBank (http://www.ncbi.nlm.nih.gov/genbank); their accession numbers are included in Figs. 5 and 6. Maximum likelihood (ML) phylogenetic analysis were implemented in MEGA6 (Tamura et al., 2013) with statistical support for tree topology obtained by bootstrapping using 1000 iterations. MEGA6 was also used to find the best fit substitution model, Tamura-Nei plus Gamma (TN93 + G) distribution for cyt-b and Tamura 3-parameters plus Gamma distribution (T92 + G) for 28 S rDNA. To build the ML topology, the nearest-neighbor-interchange (NNI) heuristic search method was used. Topologies were adjusted using Tree graph 2 software (http://treegraph.bioinfweb.info) to show only nodes with high bootstrap support.

In these trees (Figs. 5 and 6), *Triatoma* from South America were *Triatoma infestans*, *T. platensis*, *T. brasiliensis* and *T. rubrovaria*.

6.2.1. Cytogenetic data

All male *T. rubrofasciata* had a diploid chromosome complement of 25 (22 autosomes plus three sex chromosomes – X1 × 2Y) (Fig. 4). Meiotic prophase was characterized by the presence of several heterochromatic blocks scattered throughout the nucleus (Fig. 4A) with the largest chromocenter constituted by the associated sex chromosomes (arrowhead Fig. 4B). In pachytene, all autosomal bivalents exhibited a C-positive heterochromatic block at both chromosomal ends (Fig. 4B,C). The C-heterochromatin content represented about 40% of the total. The Y sex chromosome was totally heterochromatic, while the X1 and X2 sex chromosomes were the smallest of the complement (Fig. 4D). In the Brazilian specimens, the X2 chromosome was very small, sometimes difficult to detect (Fig. 4D), but in the specimens from Vietnam the X2 chromosome was much larger, similar in size to the X1 chromosome (Fig. 4D). FISH technique indicated that 45S ribosomal DNA clusters were located in the largest autosomal pair (arrowhead Fig. 4E).

6.2.2. Sequence data

From all individuals, cyt-b 667-bp and COI 606-bp fragments were obtained, without insertions or deletions (Accession numbers KR632549–KR632556). The COI sequences of the Vietnam specimens were identical, while the genetic distance between Brazilian samples was 0.5% due to 3 transition sites. The mean distance between Brazil and Vietnam sequences was 0.3%. For the cyt-b sequences, the distance between the two individuals from Brazil was 0.4% because of two transitions, and 0.2% for the two Vietnam specimens due to one transition. The mean distance between Brazil and Vietnam sequences was 0.9%, with 6 sites showing transitions.

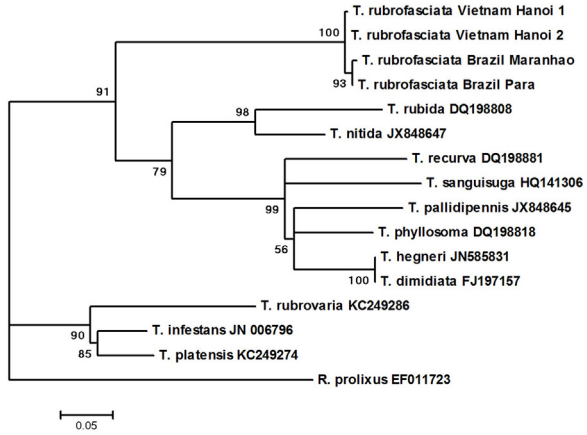

Fig. 5. Maximun likelihood phylogenetic tree inferred by Tamura-Nei plus gamma distribution model, using the 667 bp Cyt b gene fragment. Numbers on branches indicates the bootstrap with 1000 pseudoreplicates. Bar indicates the number of substitutions per site.

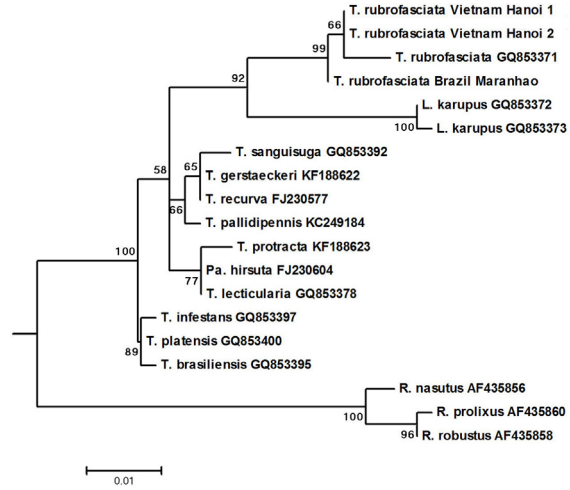

Fig. 6. Maximun likelihood phylogenetic tree inferred by Tamura 3-parameter plus gamma distribution model, using the 692 bp 28 S rDNA fragment. Numbers on branches indicates the bootstrap value with 1000 pseudoreplicates. Bar indicates the number of substitutions per site.

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For the nuclear 28S rDNA, the 692 bp fragments obtained (Accession numbers KR632546–KR632548) were manually edited to remove all ambiguously aligned nucleotide positions as is recommended for rDNA genes (Patterson and Gaunt, 2010). Vietnam individuals showed no differences, and only one variable site differentiated them from the Brazilian samples. The genetic distance was 0.0% between our *T. rubrofasciata* samples (4 specimens from Vietnam and 4 specimens from Brazil), and 0.1% from the Brazilian sample deposited in GenBank by Patterson and Gaunt (2010).

ML trees (Figs. 5 and 6) show separate clades representing South American and North American *Triatoma*, with the four *T. rubrofasciata* specimens forming a cohesive group close to the North American *Triatoma*. In the 28S rDNA tree (Fig. 6) *T. rubrofasciata* forms a clade together with *L. karupus*.

7. Discussion

7.1. Morphometric data

Morphological diversification of the Reduviidae is generally seen in obvious differences of head, thorax, legs or connexivum, much less in wing shape variation, and even less in wing venation. Contrary to the functionally significant system of the wing apparatus as a whole, wing venation has been suggested as a functionally neutral trait in insects (Grotnitsky, 1999). Its pattern shows geometric differences often needing a mathematical approach to be fully revealed, but is frequently able to distinguish cryptic species (Dujardin, 2008). In the Reduviidae, the shape of the wing does appear to show a strong phylogenetic signal. For example, the simple shape comparison of wings belonging to various genera would include the *Opisthacidius* genus within the Triatominae (Schofield and Galvão, 2009), which was also subsequently suggested by a comprehensive molecular approach (Hwang and Weirauch, 2012).

In our study, the simple comparison of wing shape disclosed some patterns in close agreement with known phylogenetic relationships. For example, there was an obvious divergence between *Ectrychotes* and the other taxa examined here. Similarly, in spite of strong morphological differences of the head and body between *Rhodnius* and *Psammolestes*, and in spite of their ecological differences, their wing venation pattern was highly similar (Fig. 3), again in complete agreement with previous phylogenetic studies (Monteiro et al., 2000; Hypsa et al., 2002). A similar close relationship was shown for the morphologically distinct *Linshcosteus* and Asiatic *Triatoma*, as previously indicated by genetic analyses (Hypsa et al., 2002; Schofield, 2004). In addition, the high wing venation similarity of the Asiatic and North American *Triatoma*, already observed for heads (Gorla et al., 1997; Patterson et al., 2001), was in complete agreement with the cyt-b and COI sequence comparison (see below). Nevertheless, the morphometric tree (Fig. 3) should only be interpreted as a phenetic dendrogram showing relative similarity between taxa. It is its comparison with previous and present molecular phylogenetic studies which suggests it also has a phylogenetic content. In the Reduviidae, wing shape similarity suggests – but does not demonstrate – common ancestry. In this regard, the greater similarity of the *Rhodnius–Psammolestes* pair, and of the *Reduvius*, *Zelurus* and *Opisthacidius* genera, with the South American *Triatoma* (Fig. 3) does not reflect known phylogenetic relationships and should be interpreted with caution.

7.2. Cytogenetic and DNA sequence data

Karyotype information is currently available for 80 Triatominae species, showing a highly conserved diploid chromosome number, ranging from 21 to 25 chromosomes in males (Panzera et al., 2010). *Triatoma rubrofasciata* is the only triatomine species with 25 chromosomes, the highest ever reported (Manna, 1950) and the only one presenting 22 autosomes. All *Triatoma* species from North America show multiple sex chromosomes (except for *T. lecticularia*), differing from the XY system reported for almost all *Triatoma* species from South America (Panzera et al., 2010).

The quantity and distribution of constitutive heterochromatin (detected by C-banding) and 45S ribosomal clusters chromosomal location (detected by FISH) are the main sources of karyotypic differentiation both at the interspecific and intraspecific level in Triatominae (Panzera et al., 2010, 2012). Here, we applied these techniques to compare chromosomal traits of *T. rubrofasciata* from Brazil and Vietnam, and their relationships with other *Triatoma* species. We confirm that, all *T. rubrofasciata* specimens here analyzed have the exceptional diploid number of 25 chromosomes, constituted by 22 autosomes plus the X1 × 2Y male sex mechanism, similar to that originally described for specimens from Calcutta, India (Manna, 1950) (Fig. 4). This author also reported that both X chromosomes were of similar size and were the smallest of the complement. Our results for specimens from Vietnam confirm these observations, but the Brazilian individuals exhibit a much smaller size of the X2 chromosomes. Size variation of sex chromosomes has been noted in populations of other triatomine species such as *T. infestans*, and seems to reflect a geographical differentiation occurring during the dispersal of these species (Panzera et al., 2004). Furthermore, we described, for the first time, that all *T. rubrofasciata* specimens have a high C-heterochromatin content as well as the autosomal location of the ribosomal clusters (Fig. 4). This high C-heterochromatin content and its chromosomal distribution are also similar to that observed in North American *Triatoma* species such as *T. protracta*, *T. barberi* and *T. lecticularia* (Panzera et al., 2010; 2012). All these chromosomal features suggest a chromosomal identity between *T. rubrofasciata* of Brazil and Vietnam as well as a close evolutionary relationship between *T. rubrofasciata* and North America *Triatoma* species.

DNA sequence information about *T. rubrofasciata* is scarce. Although several genes are available in GenBank (16S, 18S, 28S, COI, COII), none of the previous phylogenetic analyses included simultaneously specimens from the Old and New Worlds (e.g., Hypsa et al., 2002; Patterson and Gaunt, 2010). In this paper we compare DNA sequence data between Asiatic and New World specimens of *T. rubrofasciata*, fully confirming their specific identity. ML trees recovered all *T. rubrofasciata* individuals forming a cohesive clade for all three genes analyzed (Figs. 5 and 6). As suggested by former analyses (Hypsa et al., 2002; Patterson and Gaunt, 2010), our results confirm the close association between *T. rubrofasciata* with the *Linshcosteus* genus (Fig. 6), and between the Asiatic and North American clades of *Triatoma* (Figs. 5 and 6). Moreover, the comparative analyses of the nuclear (28S) and mitochondrial (cyt-b and COI) sequences showed no evidence of incipient speciation between Asiatic and New World specimens of *T. rubrofasciata*. On the contrary, their high genetic similarity suggests either a very recent dispersal or an ongoing exchange between the continents.

In conclusion, our chromosome and molecular data suggest that *T. rubrofasciata* individuals from Asia and South America represent conspecific populations. And while our cytogenetic and gene sequence results, like morphometric data, clearly indicate that the Triatomini form two main groups corresponding to North and South America, the comparative analyses of nuclear (28S) and mitochondrial (cyt-b and COI) sequences confirm the close association between the Asiatic and North American *Triatoma* (Figs. 5 and 6).

7.3. Concluding assessment

Our assessment of available information seems to support the idea that the Asian species of Triatominae – *Triatoma* and *Linshcosteus* – form a closely-related group within the Triatomini, and do

not merit the concept of separate tribal status for *Linshcosteus* as proposed by Carcavallo et al. (2000). Moreover, this group appears most closely associated with the North American clade of Triatomini, rather than with the South American clade, as shown by wing morphometric, cytogenetic, and sequence comparisons. This suggests – as also indicated by its association with shiprats, and reported occurrence in so many port areas of the tropics and subtropics – that *T. rubrofasciata* originated in North America, and was subsequently carried elsewhere on ships.

Although historically there appear to have been three main seaborne trade routes from the Americas (Roullier et al., 2013) we can suggest that the origin of the Asiatic Triatominae would most likely have been the eastern seaboard of North America – rather than the western – because, except for the Hawaiian islands, there are as yet no reports of *T. rubrofasciata* from any of the Pacific ports of the Americas. This could indicate that the original emigration of these insects would have been associated with Atlantic trade routes beginning in the 16th century (cf. Roullier et al., 2013); yet they were already seen to be problematic in parts of Asia by the 18th century, with new species being described by the end of the 19th. Given the relatively slow pace of seaborne communication in those days, this is both a rapid geographical spread, and a rapid rate of morphological divergence.

Although recorded from port areas however, there are no reports of silvatic populations of *T. rubrofasciata* in North America, that could have indicated their point of origin. By contrast, both *T. sanguisuga* and *T. lecticularia* are widespread in the southern USA, both in peridomestic and silvatic populations, particularly in the region of Texas, Georgia, Florida, and the Carolinas. They are morphologically similar to *T. rubrofasciata*, and the three have been synonymised at various times [to quote Lent and Wygodzinsky (1979): "In 1872, Stal synonymised his *lecticularius* with *variegatus*, a synonym of *rubrofasciata*. Neiva (1914) placed *lecticularius* in the synonymy of *sanguisuga*. . ."]. The original description of *lecticularius* was based on two specimens labelled as being from "Carolina" and "India orientalis". The Carolina specimen had been synonymised with *T. sanguisuga*, while the other – by dismissing the label as "obviously erroneous" – was designated the lectotype1 of *T. lecticularia* by Lent and Wygodzinsky (1979). But if the original label was not erroneous (which we doubt) then this lectotype of *T. lecticularia* may in fact be a *T. rubrofasciata*.

The taxonomic confusion seems to reflect the morphological similarities of these three taxa, all of which show overlapping population variability. Combined with their morphometric and cytogenetic similarity, we suggest that these three taxa are very closely related and probably share a common ancestral form from the area that is now the south eastern USA, which also represents the origin of the Asiatic Triatominae. The South American populations of *T. rubrofasciata* may derive directly from this source but, given the continuous maritime transportation, they might also contain insects carried back from Asia to Latin America.

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